

Requirement Analysis

Clinical Specialist - Analytics Dashboard Journey

Metric	Details
Project Title	Predictive Risk Clustering and Visual Informatics for Cardiovascular Disease
Repository	Heart-Disease-Analysis
Website	Flask web application with an embedded Tableau Dashboard and Tableau Story
Team Members	K. Bhargav

Project Description

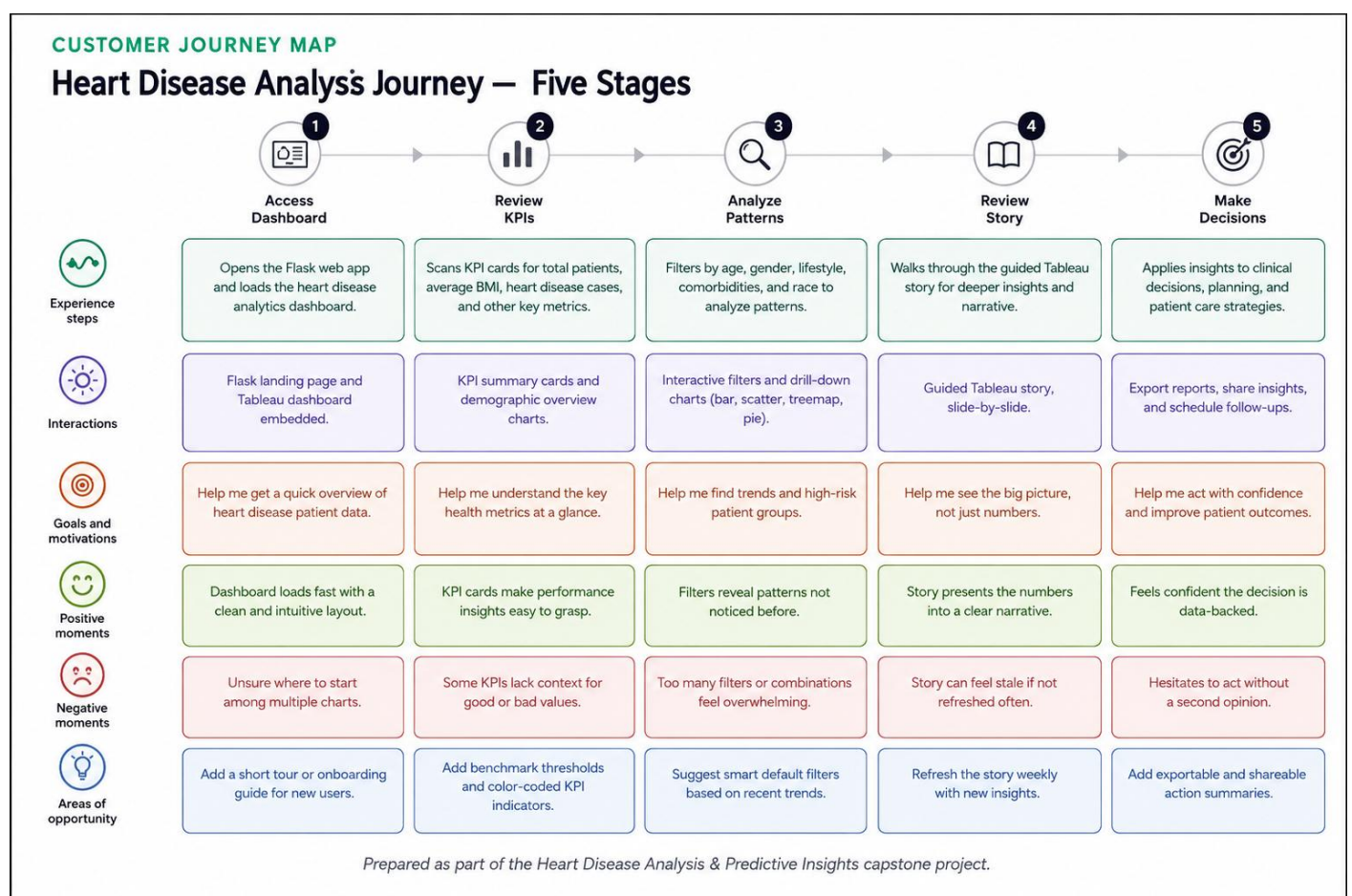
The project analyzes approximately 4,494 complete patient diagnostic records across 18 clinical variables using Tableau Public. It delivers critical healthcare insights through three comprehensive interactive Tableau Dashboards and an aligned 7-part Tableau Data Story, both embedded inside a custom-engineered, lightweight Flask-based web application framework.

The embedded portal platform covers the following specific visualization environments derived directly from active system dashboards:

- **Dashboard 1: Baseline Demographics & Clinical Factors Matrix**
 - Gender vs. Heart Disease distribution trends.
 - Precise Age Bracket distribution tracking counts from 18 to 80 or older.
 - Diabetic status mapping cross-referenced directly with historical Stroke counts.
 - Dual-behavioral impact metrics mapping Smoking and Alcohol consumption frequencies against case volumes.
 - **Dashboard 2: Secondary Co-morbidities & Lifestyle Metrics**
 - Other Health Diseases vs. Stroke matrix tracking joint occurrences of Asthma, Kidney Disease, and Skin Cancer.
 - Race-wise Heart Disease distribution pie charts showing statistical prevalence metrics (e.g., White: 75.57%, Black: 17.81%, Hispanic: 3.73%, Asian: 2.07%).
 - General Health self-reports bubble clusters showing absolute patient volumes (Poor: 92, Fair: 140, Very Good: 67, Good: 172).
 - Physical Activity impact charts showing split behavioral trends (Active: 279, Sedentary: 204).
 - **Dashboard 3: Deep Metabolic Risk Analysis**
 - Comprehensive Age vs. Body Mass Index (BMI) vs. Diabetes tree-map blocks mapping the critical distribution scale between a baseline BMI of 23.91 and a high-risk severe cluster threshold of 42.78.
 - Triple-axis patient summaries mapping Stroke vectors for patients concurrently suffering from advanced Diabetes and underlying Heart Diseases.
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Clinical Customer Persona

- **Primary User:** Diagnostic Health Specialist / Cardiovascular Informatics Consultant
- **Age Bracket:** 30–50 years
- **Occupation:** Clinical Care Consultant
- **Core Experience:** Manages daily patient validation workflows, diagnostic tracking metrics, and preventative medical risk evaluations.
- **System Goals:**
 - Understand overlapping patient risk behaviors and co-morbidities dynamically.
 - Monitor active clinical indicators and key performance indicators cleanly without data cognitive overload.
 - Identify high-risk population clusters and age cohort distribution trends at a glance.
 - Accelerate clinical validation timelines and optimize preventative care workflows.
 - Make data-backed decisions with complete configuration confidence.
- **Operational Challenges:**
 - Massive influx of patient diagnostic datasets containing thousands of rows and rows of unorganized parameters.
 - Cognitive visual overload when attempting to cross-reference multiple health factors manually from tabular text spreadsheets.
 - Rigid digital systems that completely lack simultaneous dual-axis summaries or interactive cloud filters.
 - Cluttered, generic UI templates that obscure medical insights under placeholder text strings.



Customer Journey Map Matrix

Dashboard Analytics Journey: Five Lifecycle Stages

Phase Metric	1. Access Dashboard	2. Review KPIs	3. Analyze Behaviour	4. Review Story	5. Make Decisions
Experience Steps	Opens the local Flask application and loads the main health portal page workspace.	Scans integrated top dashboards for baseline demographic and gender counts.	Utilizes interactive parameters grids to isolate Smoking, Alcohol, and Diabetic charts.	Navigates through the 7part Tableau story narrative layout step by slide.	Applies hidden risk trend insights to patient care, treatment plans, and research workflows.
System Interactions	Custom Flask workspace web application UI home path and core HTML cards layout.	KPI dashboard summaries container box with accent borders.	Live embedded cloud cross-filters, checkboxes, pie charts, and scatter plots.	Sandboxed Tableau Public Storyboard navigation interface framework.	Clinical consultation sessions and exportable informatics logs.
Phase Metric	1. Access Dashboard	2. Review KPIs	3. Analyze Behaviour	4. Review Story	5. Make Decisions
Goals & Motivations	Help me get a fast performance snapshot of the incoming diagnostic data queue.	Help me understand what the patient volume metrics and averages indicate instantly.	Help me locate latent pathological patterns that are worth acting upon immediately.	Help me see the comprehensive clinical big picture, not just disconnected separate charts.	Help me act with complete confidence when diagnosing compounding risk factors.

Positive Moments	Application engine loads rapidly from scratch with zero template text clutter.	Core metrics layout values are incredibly easy to grasp and scan visually.	Filtering elements expose overlapping risk connections that were completely hidden before.	The structured narrative sequence binds separate datasets into one clear story conclusion.	Feels fully confident that final medical treatment strategies are entirely databacked.
Negative Moments	Unsure where to focus initial attention due to multiple interactive layout blocks.	Certain static raw parameters might lack contextual historical baselines at a glance.	Too many dynamic factor combinations can initially feel confusing to read.	Storyboards can feel static if the underlying data framework isn't updated frequently.	Hesitates to lock down final diagnostic outcomes without a secondary peer review.
Areas of Opportunity	Integrate a brief visual onboarding layout guide block on the interface.	Add colorcoded warning alert thresholds to high-risk numerical parameters cards.	Provide optimized default filter values based on high-risk medical age cohorts.	Refresh dashboard connection links frequently with active real-time variables.	Add exportable, shareable PDF clinical summary document generation triggers.

Prepared as part of the Predictive Risk Clustering and Visual Informatics for Cardiovascular Heart Disease Analysis capstone project.